Digital Twins for Smart Cities: A Bibliometric Analysis of the Decision Support Gap

ADM 804 — COPPEAD/UFRJ

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1 Visão Geral do Corpus

Table 1: Table 1 — General statistics of the bibliometric corpus

Métrica	Valor
Total de artigos	159
Período	2020 – 2026
Periódicos únicos	67
Autores únicos	656
Países	10
Média de citações por artigo	28
Mediana de citações	4
Total de citações	4456

2 Crescimento da Área

2.1 Publicações por ano

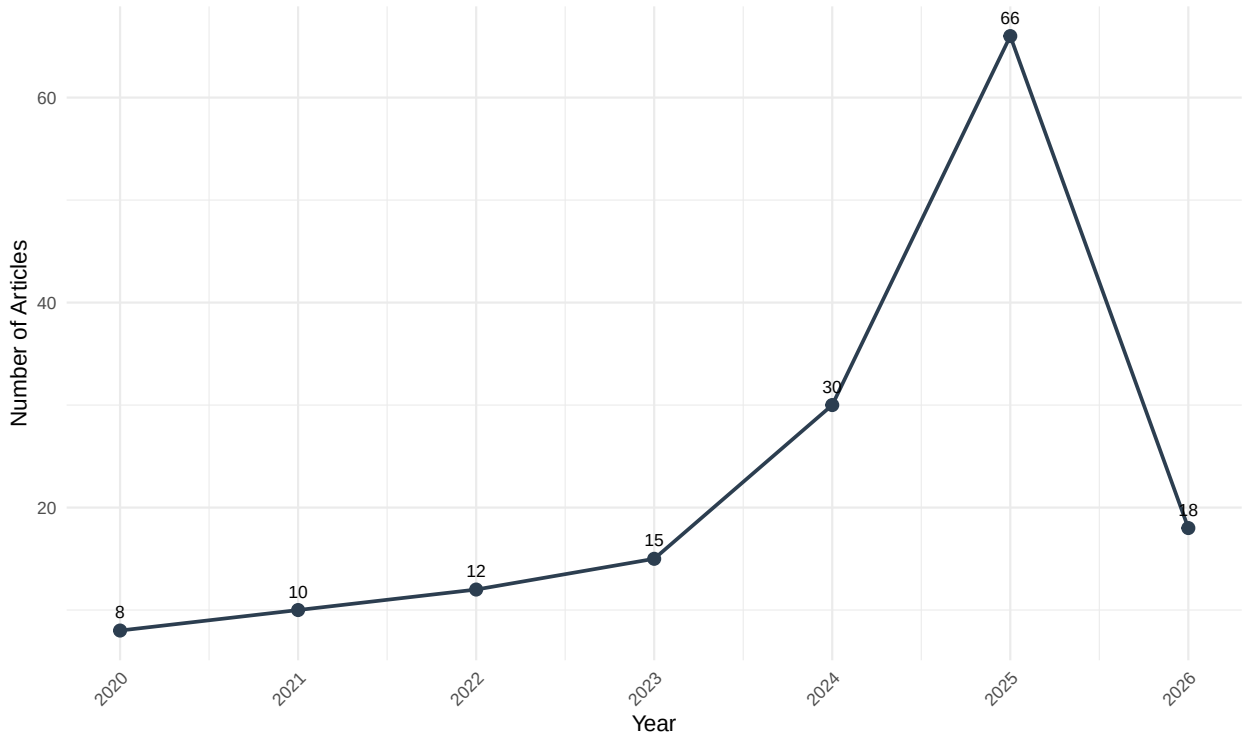

Figure 1: Annual scientific production on digital twins and smart cities

3 Autores Mais Prolíficos

Table 2: Table 2 — Top 10 most prolific authors

Author	Articles
LIU Y	4
COELHO M	3
KIM J	3
LI H	3
LI X	3
LI Y	3
YE X	3
AKROYD J	2
ALLAN M	2
BALLOT E	2

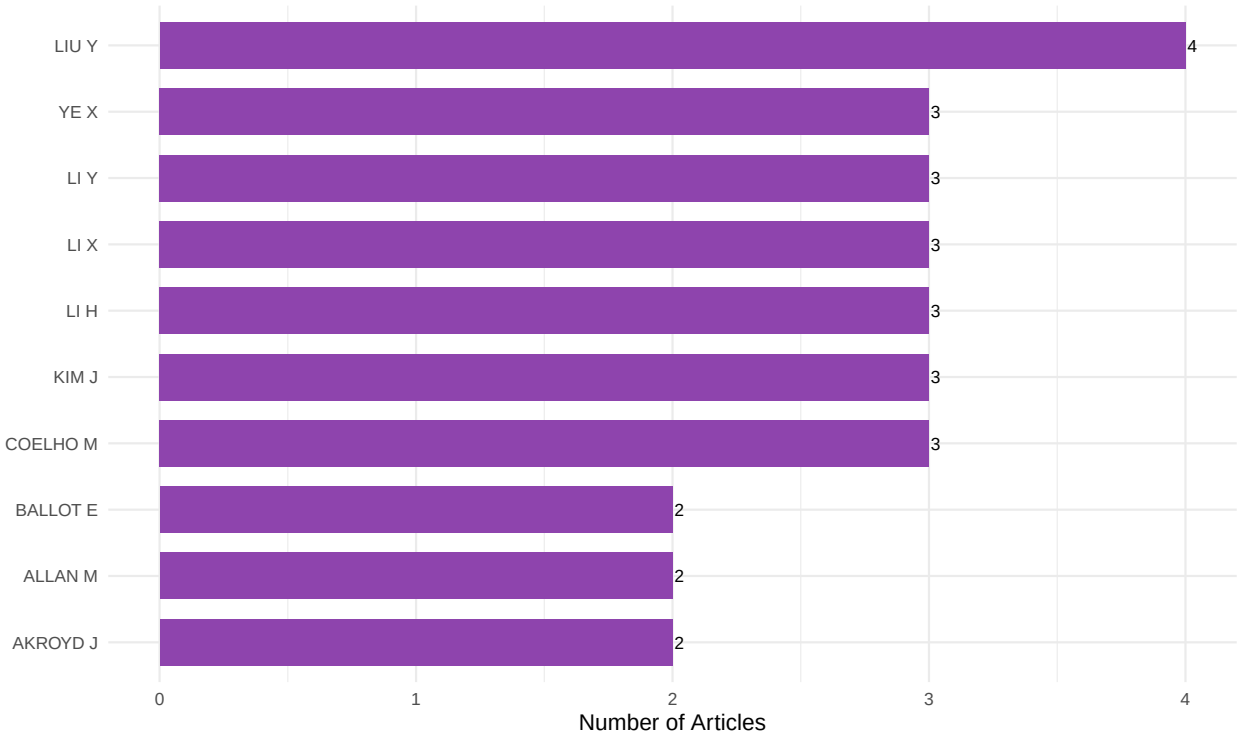

Figure 2: Top 10 most prolific authors

4 Evolução Temporal dos Autores

Este gráfico mostra a produção dos autores mais prolíficos ao longo do tempo. Cada ponto representa um artigo; o tamanho do ponto é proporcional ao número de citações recebidas.

Table 3: Table 3 — Author production over time (selected entries)

Author	Year	Title	Journal	DOI	TC	TC/Y
LIU Y	2026	LIFE CYCLE THINKING IN CULTURAL HERITAGE: SYSTE...	JOURNAL OF GREEN BUILDING	NA	0	0.000000
LIU Y	2026	LIFE CYCLE THINKING IN CULTURAL HERITAGE: SYSTE...	JOURNAL OF GREEN BUILDING	10.3992/jgb.21.1.367	0	0.000000
LIU Y	2025	FEDERATED DIGITAL TWINS PLATFORM FOR SMART CITY...	INTERNATIONAL JOURNAL OF PR...	10.1016/j.ijpe.2025.109772	0	0.000000
LIU Y	2021	DIGITAL TWIN-DRIVEN APPROACH FOR SMART CITY LOG...	ADVANCES IN PRODUCTION MANA...	10.1007/978-3-030-85910-7_25	14	2.333333
LI H	2026	DIGITAL-TWIN-DRIVEN URBAN LIFECYCLE PARADIGM: E...	FRONTIERS IN SUSTAINABLE CI...	10.3389/frsc.2026.1733281	0	0.000000
KIM J	2025	ANALYSIS OF URBAN MANAGEMENT INFORMATION TECHNO...	INTERNATIONAL JOURNAL OF UR...	10.1080/12265934.2025.2452501	3	1.500000
LI Y	2025	SPATIO-TEMPORAL DATA FUSION TECHNIQUES FOR MODE...	GEO-SPATIAL INFORMATION SCI...	10.1080/10095020.2024.2350175	18	9.000000
LI H	2025	AI-ENABLED DIGITAL TWIN FRAMEWORK FOR SAFE AND ...	SUSTAINABILITY	10.3390/su17104391	3	1.500000
LI X	2025	AI-ENABLED DIGITAL TWIN FRAMEWORK FOR SAFE AND ...	SUSTAINABILITY	10.3390/su17104391	3	1.500000
YE X	2025	TOWARD ETHICAL GEODESIGN IN THE URBAN DIGITAL T...	JOURNAL OF PLANNING EDUCATI...	10.1177/0739456X251359233	1	0.500000
YE X	2025	INTEGRATING URBAN DIGITAL TWIN WITH CLOUD-BASED...	JOURNAL OF PLANNING EDUCATI...	10.1177/0739456X251316185	6	3.000000
YE X	2025	INTEGRATING SPATIOTEMPORAL VISION TRANSFORMER I...	JOURNAL OF PLANNING EDUCATI...	10.1177/0739456X251391121	2	1.000000
LI X	2025	SMART COMMUNITY-DRIVEN SUSTAINABLE URBAN TRANSI...	JOURNAL OF DATA AND INFORMA...	10.2478/jdis-2025-0037	0	0.000000
LI X	2025	RESEARCH ON THE INFRASTRUCTURE RESILIENCE SYSTE...	SUSTAINABILITY	10.3390/su17188232	0	0.000000
LI H	2025	RESEARCH ON THE INFRASTRUCTURE RESILIENCE SYSTE...	SUSTAINABILITY	10.3390/su17188232	0	0.000000

5 Centralidade dos Autores na Rede de Co-autoria

A centralidade mede a importância de cada autor na rede de colaboração. Autores com alta *betweenness centrality* são pontes entre grupos de pesquisa; autores com alto *degree* têm muitos co-autores diretos.

Table 4: Table 4 — Top 15 authors by betweenness centrality in the co-authorship network

	Author	Degree	Betweenness	Closeness	PageRank
CHEN S	CHEN S	20	0.0017	0.5833	0.0031
LI H	LI H	14	0.0011	0.5316	0.0025
LI X	LI X	17	0.0011	0.5316	0.0030
ZHU D	ZHU D	7	0.0009	0.4200	0.0017
LI Y	LI Y	12	0.0007	0.4286	0.0024
HUANG Z	HUANG Z	16	0.0007	0.5250	0.0027
LU Q	LU Q	4	0.0005	0.3684	0.0014

Table 4: Table 4 — Top 15 authors by betweenness centrality in the co-authorship network

	Author	Degree	Betweenness	Closeness	PageRank
COELHO M	COELHO M	16	0.0003	1.0000	0.0031
YE X	YE X	13	0.0003	0.8824	0.0033
GADEKALLU T	GADEKALLU T	14	0.0002	1.0000	0.0027
THUVANDER L	THUVANDER L	13	0.0002	1.0000	0.0027
WANG Y	WANG Y	13	0.0002	0.8421	0.0025
ZHANG F	ZHANG F	13	0.0002	0.8421	0.0025
LIU Y	LIU Y	10	0.0001	0.6667	0.0028
BIBRI S	BIBRI S	9	0.0001	1.0000	0.0026

6 Top Periódicos

Table 5: Table 5 — Top 10 most productive journals

Journal	Articles
SUSTAINABILITY	20
SUSTAINABLE CITIES AND SOCIETY	16
SMART CITIES	13
CITIES	8
URBAN SCIENCE	8
WATER	6
ENVIRONMENT AND PLANNING B-URBAN ANALYTICS AND CITY SCIENCE	5
ISPRS INTERNATIONAL JOURNAL OF GEO-INFORMATION	4
JOURNAL OF MANAGEMENT IN ENGINEERING	3
JOURNAL OF PLANNING EDUCATION AND RESEARCH	3

7 Top Países Produtores

Table 6: Table 6 — Top 10 countries by scientific production

Country	Articles
CHINA	23
USA	18
ITALY	16
UNITED KINGDOM	11
AUSTRALIA	7
FRANCE	7
IRAN	7
SPAIN	7
GERMANY	5

Table 6: Table 6 — Top 10 countries by scientific production

Country	Articles
NETHERLANDS	5

8 Top Artigos Mais Citados

Table 7: Table 7 — Top 10 most cited articles

	Author	Title	Journal	Year	Citations
DENG T, 2021, J MANAG SCI ENG	DENG T et al.	A SYSTEMATIC REVIEW OF A DIGITAL TWIN CITY: A NEW PATTERN OF URBAN ...	JOURNAL OF MANAGEMENT SCIENCE AN...	2021	517
DEMBSKI F, 2020, SUSTAINABILITY	DEMBSKI F et al.	URBAN DIGITAL TWINS FOR SMART CITIES AND CITIZENS: THE CASE STUDY O...	SUSTAINABILITY	2020	358
SCHROTTER G, 2020, PFG-J PHOTOGRAMM REMOTE SENS GEOINF SCI	SCHROTTER G et al.	THE DIGITAL TWIN OF THE CITY OF ZURICH FOR URBAN PLANNING	PFG-JOURNAL OF PHOTOGRAMMETRY RE...	2020	340
ALLAM Z, 2022, SMART CITIES	ALLAM Z et al.	THE METAVERSE AS A VIRTUAL FORM OF SMART CITIES: OPPORTUNITIES AND ...	SMART CITIES	2022	312
WEIL C, 2023, SUSTAINABLE CITIES SOC	WEIL C et al.	URBAN DIGITAL TWIN CHALLENGES: A SYSTEMATIC REVIEW AND PERSPECTIVES...	SUSTAINABLE CITIES AND SOCIETY	2023	217
KETZLER B, 2020, BUILT ENVIRON	KETZLER B et al.	DIGITAL TWINS FOR CITIES: A STATE OF THE ART REVIEW	BUILT ENVIRONMENT	2020	209
RAMU S, 2022, SUST CITIES SOC	RAMU S et al.	FEDERATED LEARNING ENABLED DIGITAL TWINS FOR SMART CITIES: CONCEPTS...	SUSTAINABLE CITIES AND SOCIETY	2022	170
NOCHTA T, 2021, J URBAN TECHNOL	NOCHTA T et al.	A SOCIO-TECHNICAL PERSPECTIVE ON URBAN ANALYTICS: THE CASE OF CITY-...	JOURNAL OF URBAN TECHNOLOGY	2021	156
FERRE-BIGORRA J, 2022, CITIES	FERRE-BIGORRA J et al.	THE ADOPTION OF URBAN DIGITAL TWINS	CITIES	2022	150

HAM Y, 2020, J MANAGE ENG	HAM Y et al.	PARTICIPATORY SENSING AND DIGITAL TWIN CITY: UPDATING VIRTUAL CITY ...	JOURNAL OF MANAGEMENT IN ENGINEE...	2020	141

9 Rede de Co-autoria

10 Rede de Colaboração entre Países

11 Author Coupling

O acoplamento bibliográfico (bibliographic coupling) conecta autores que citam as mesmas referências, revelando frentes de pesquisa com base intelectual comum.

```
[1] "Matrix is empty!!"
Coupling error: non-conformable arguments
Bibliographic coupling could not be computed.
```

12 Contagem de Palavras-chave

Table 8: Table 8 — Top 20 author keywords

Keyword	Freq
digital twin	68
smart city	27
digital twins	22
smart cities	19
urban planning	16
sustainability	13
urban digital twin	12
machine learning	9
urban	8
urban digital twins	8
artificial intelligence	6
metaverse	6
simulation	6
urban management	6
data	5
decision support	5
sustainable development	5
urban governance	5
analysis	4
built environment	4

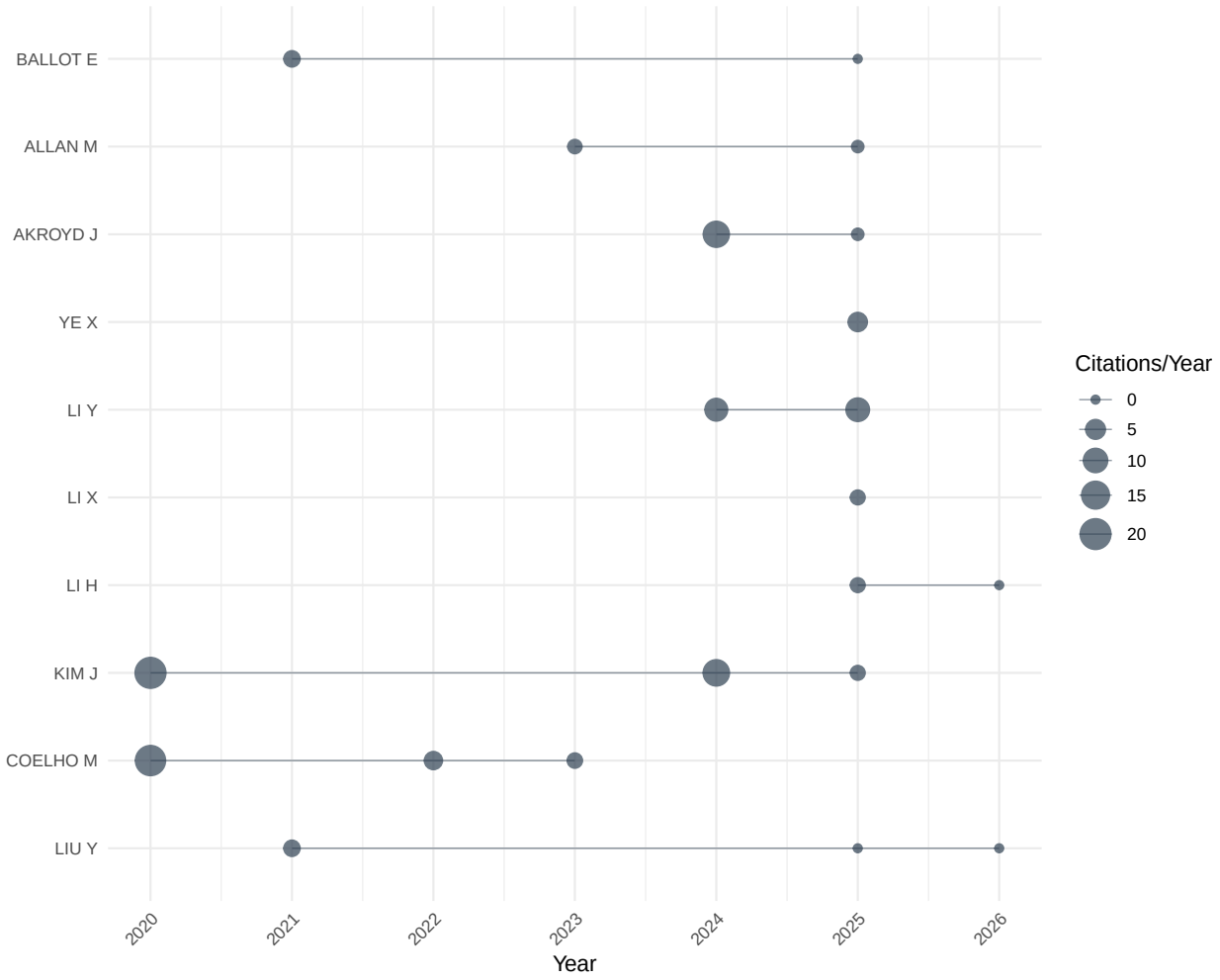

Figure 3: Author production over time — top 10 authors (point size proportional to citations)

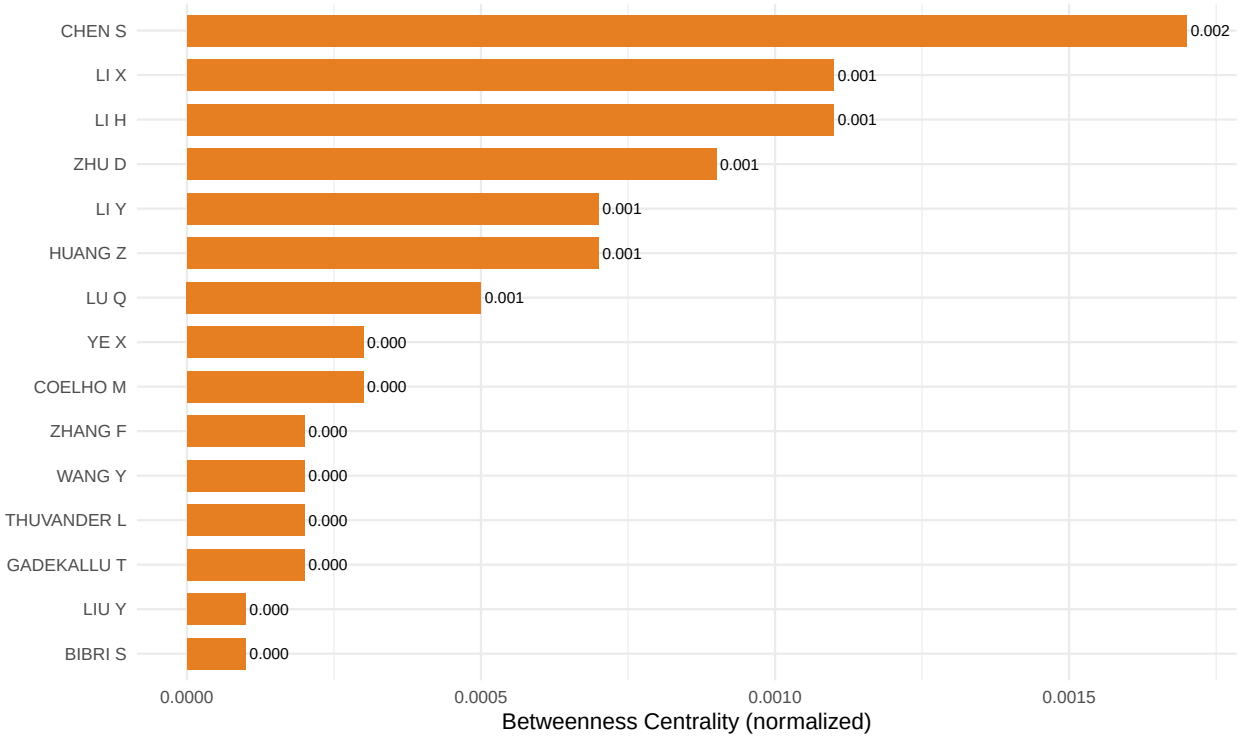

Figure 4: Top 15 authors by betweenness centrality

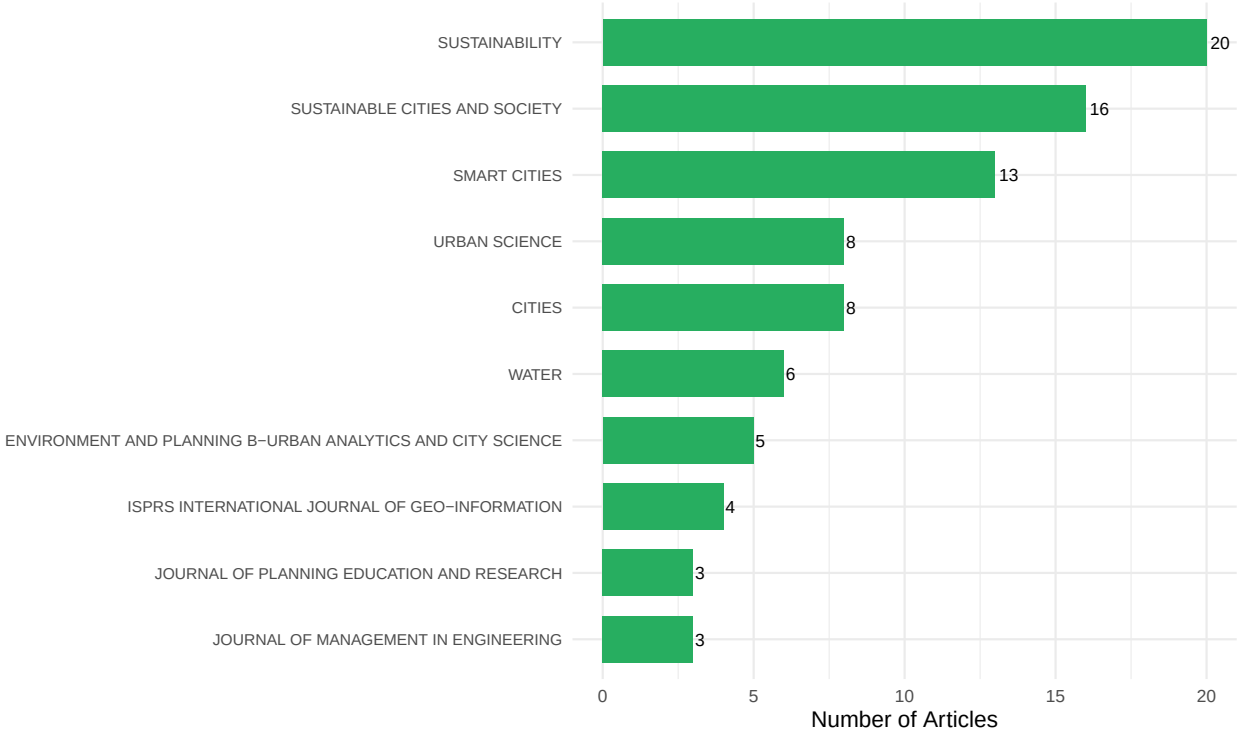

Figure 5: Top 10 most productive journals

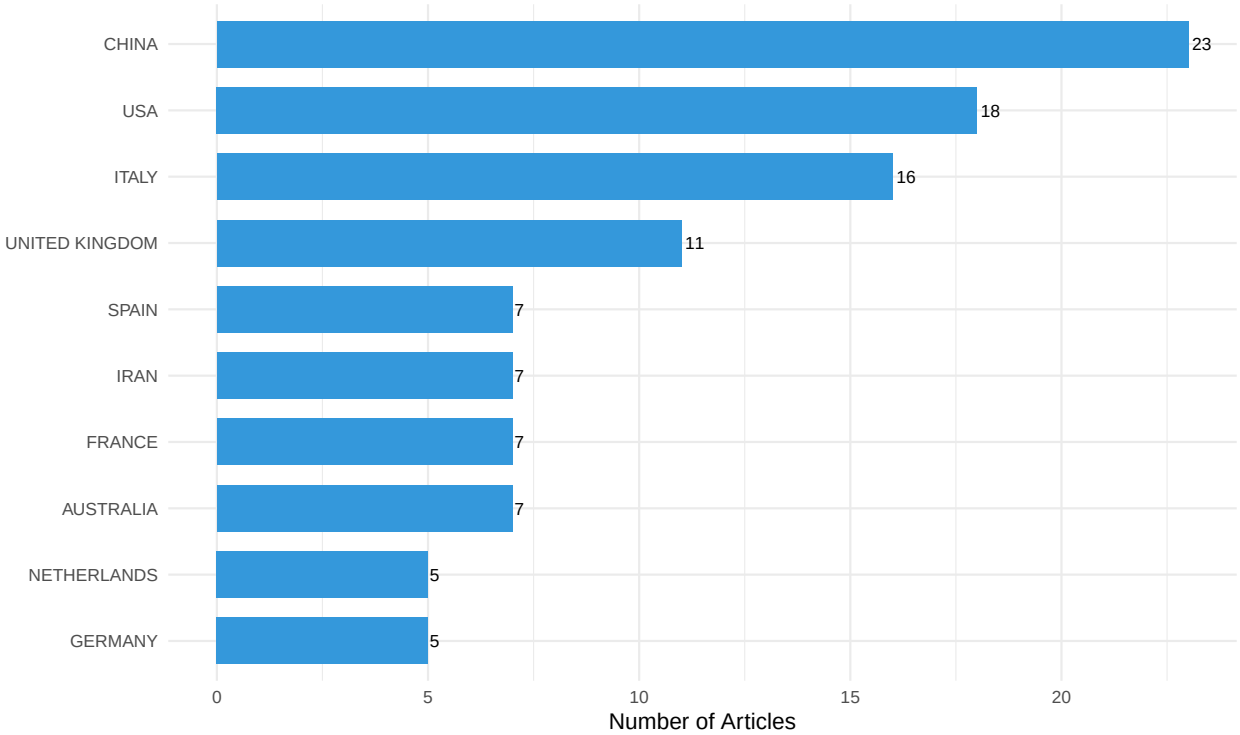

Figure 6: Top 10 countries by scientific production

13 Wordcloud de Palavras-chave

14 Rede de Co-ocorrência de Palavras-chave

15 Mapa Temático

O mapa temático posiciona os clusters de palavras-chave em quatro quadrantes:

- **Motor themes** (alta centralidade + alta densidade): temas centrais e bem desenvolvidos
- **Niche themes** (baixa centralidade + alta densidade): temas periféricos mas coesos
- **Basic themes** (alta centralidade + baixa densidade): temas centrais mas pouco desenvolvidos
- **Emerging/declining themes** (baixa centralidade + baixa densidade): temas periféricos e pouco desenvolvidos

Table 9: Table 9 — Thematic clusters from the strategic diagram

Cluster	Centrality	Density	Quadrant	Keywords
digital twin	9	7	Motor	digital twin 67

smart city 27 urban planning 16 urban 8 simulation 6 urban management 6 decision support 5 urban governance 5 built environment 4 iot 4 | |digital twins | 6| 5|Motor |digital twins 22 smart cities 19 internet of things 4 ontology 3 stakeholder engagement 3 climate change 2 learning 2 machine

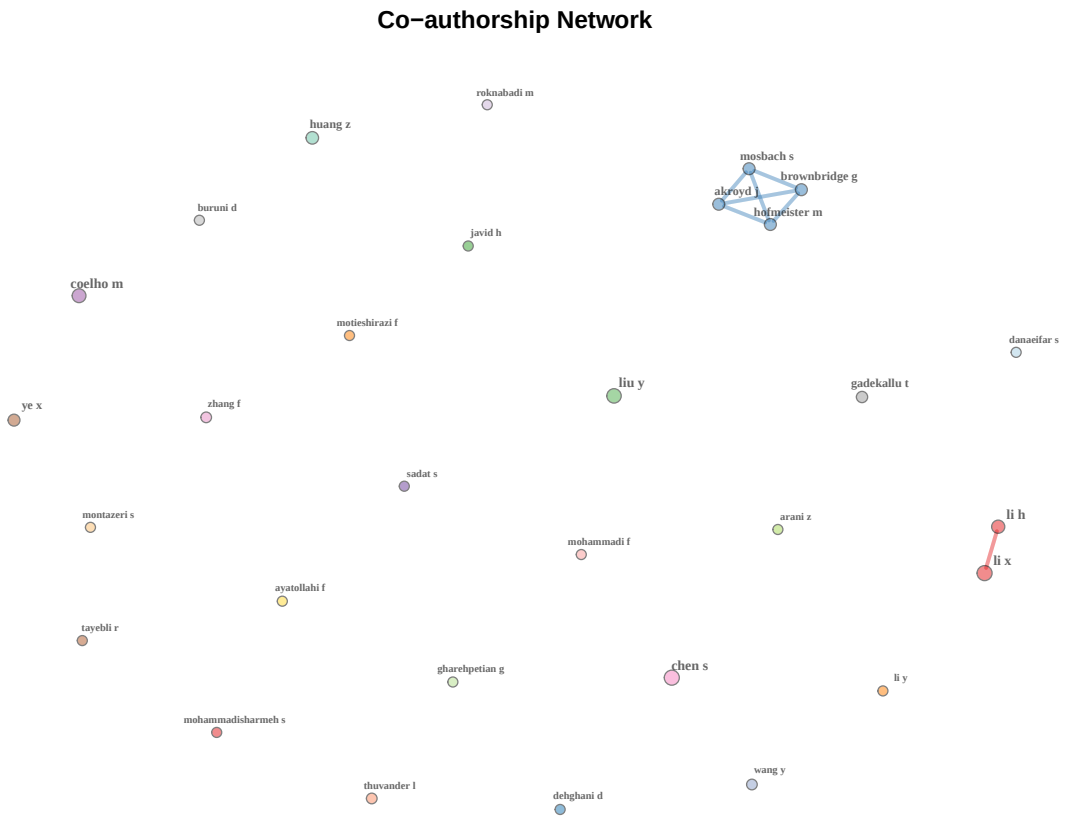

Figure 7: Co-authorship network — clusters represent research groups

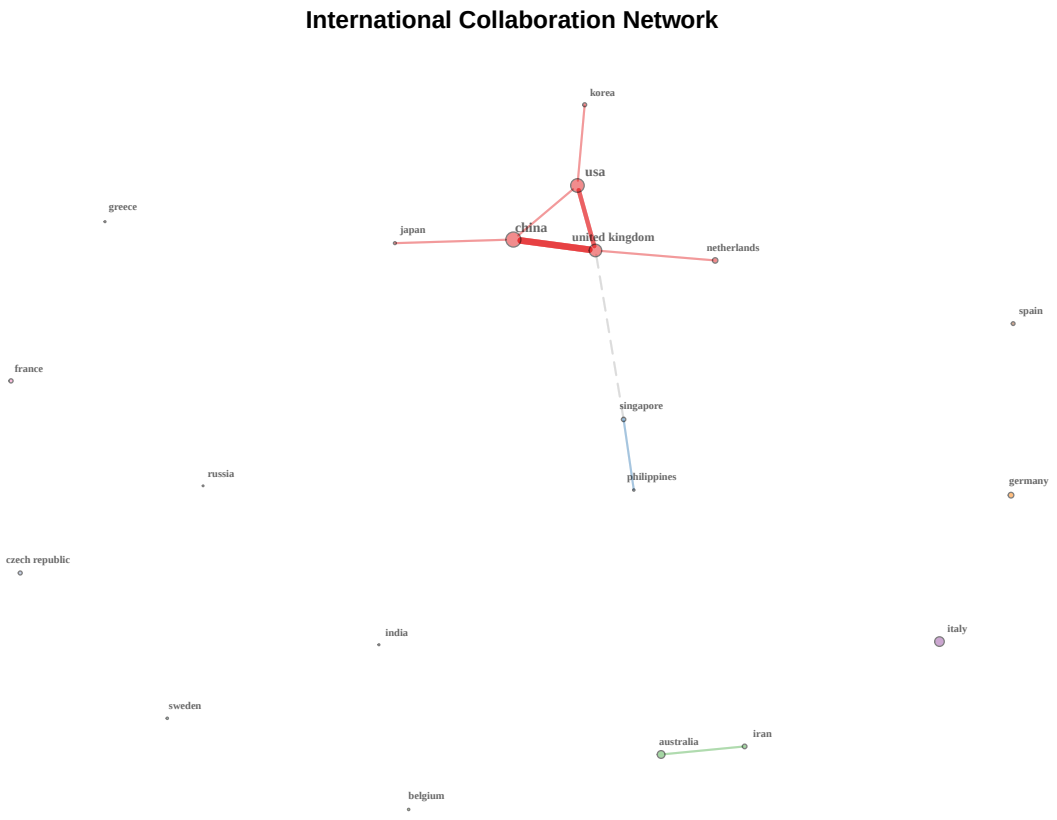

Figure 8: International collaboration network between countries

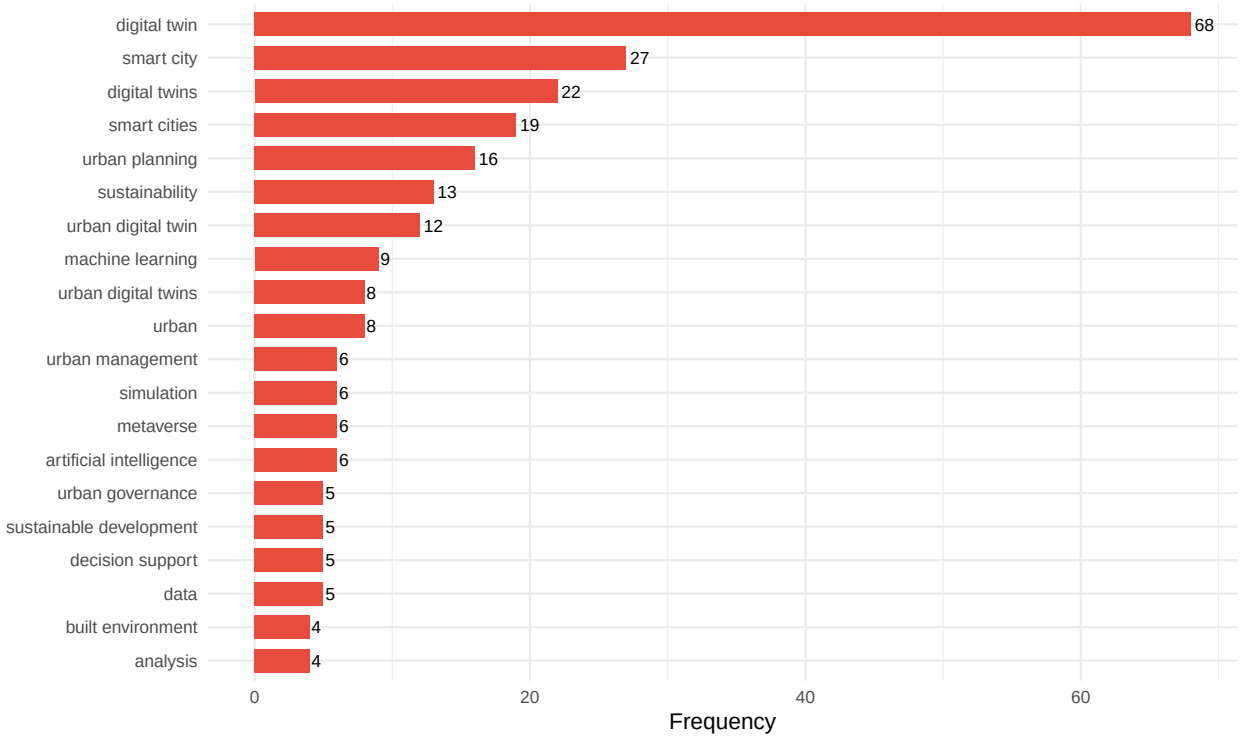

Figure 9: Top 20 author keywords by frequency

2 open data 2 quality of life 2 | |sustainability | 7| 1|Motor |sustainability 13 urban digital twin 12 data 5 analysis 4 cities 4 energy transition 3 smart 3 climate neutrality 2 technology 2 urban climate 2 | |machine learning | 8| 6|Motor |machine learning 9 artificial intelligence 6 planning 4 bibliometric analysis 3 decision-making 3 deep learning 3 participatory planning 3 decision support systems 2 digital twin technology 2 human thermal comfort 2 | |urban digital twins | 3| 2|Motor |urban digital twins 8 citizen participation 4 digitalization 3 5g 2 city planning 2 sustainable smart cities 2 | |metaverse | 5| 4|Motor |metaverse 6 internet of things (iot) 4 architecture 3 artificial 3 gis 3 digital twin city 2 intelligence 2 | |sustainable development | 4| 3|Motor |sustainable development 5 circular economy 4 cultural heritage 3 life cycle assessment 3 urban regeneration 3 china 2 graph neural networks 2 | |digital twin (dt) | 1| 9|Motor |digital twin (dt) 3 federated learning 2 air quality 2 autonomous vehicles 2 intelligent transportation systems 2 making 2 resource 2 traffic congestion 2 traffic management 2 | |campus | 2| 8|Motor |campus 2 geographic information system 2 living lab 2 |

16 Evolução Temática

17 Análise do Gap — Suporte à Decisão

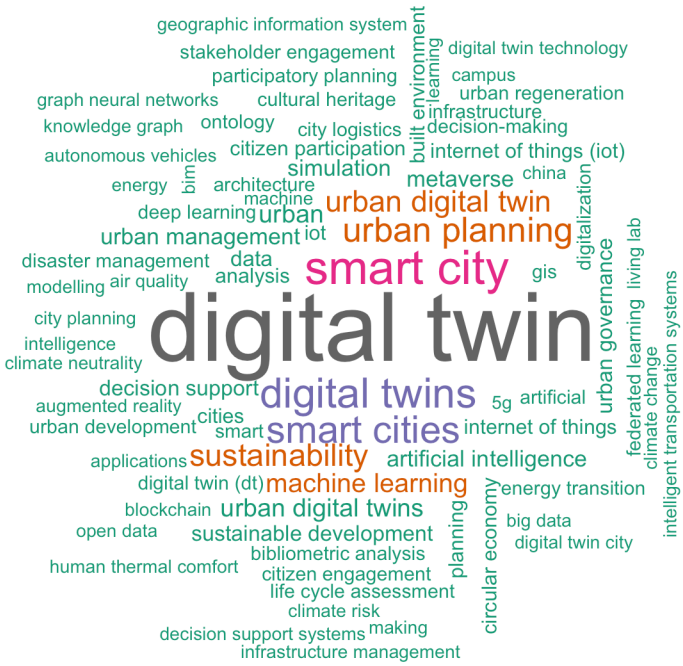

Figure 10: Word cloud of author keywords

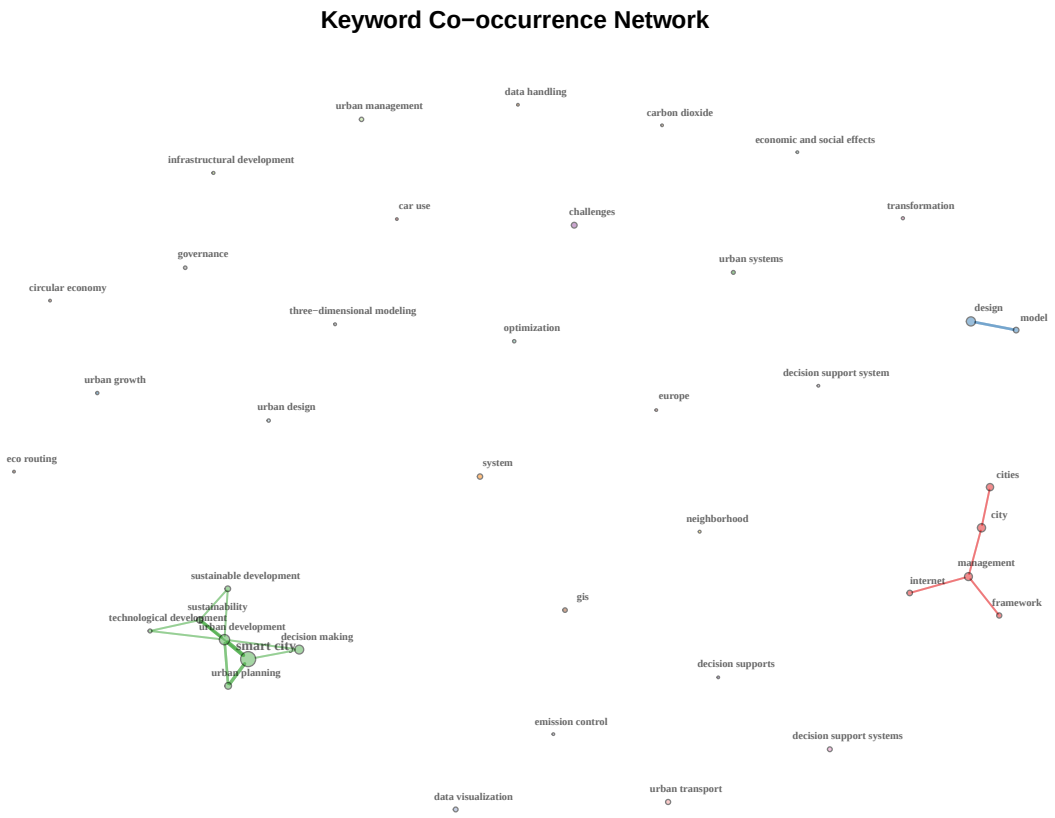

Figure 11: Keyword co-occurrence network

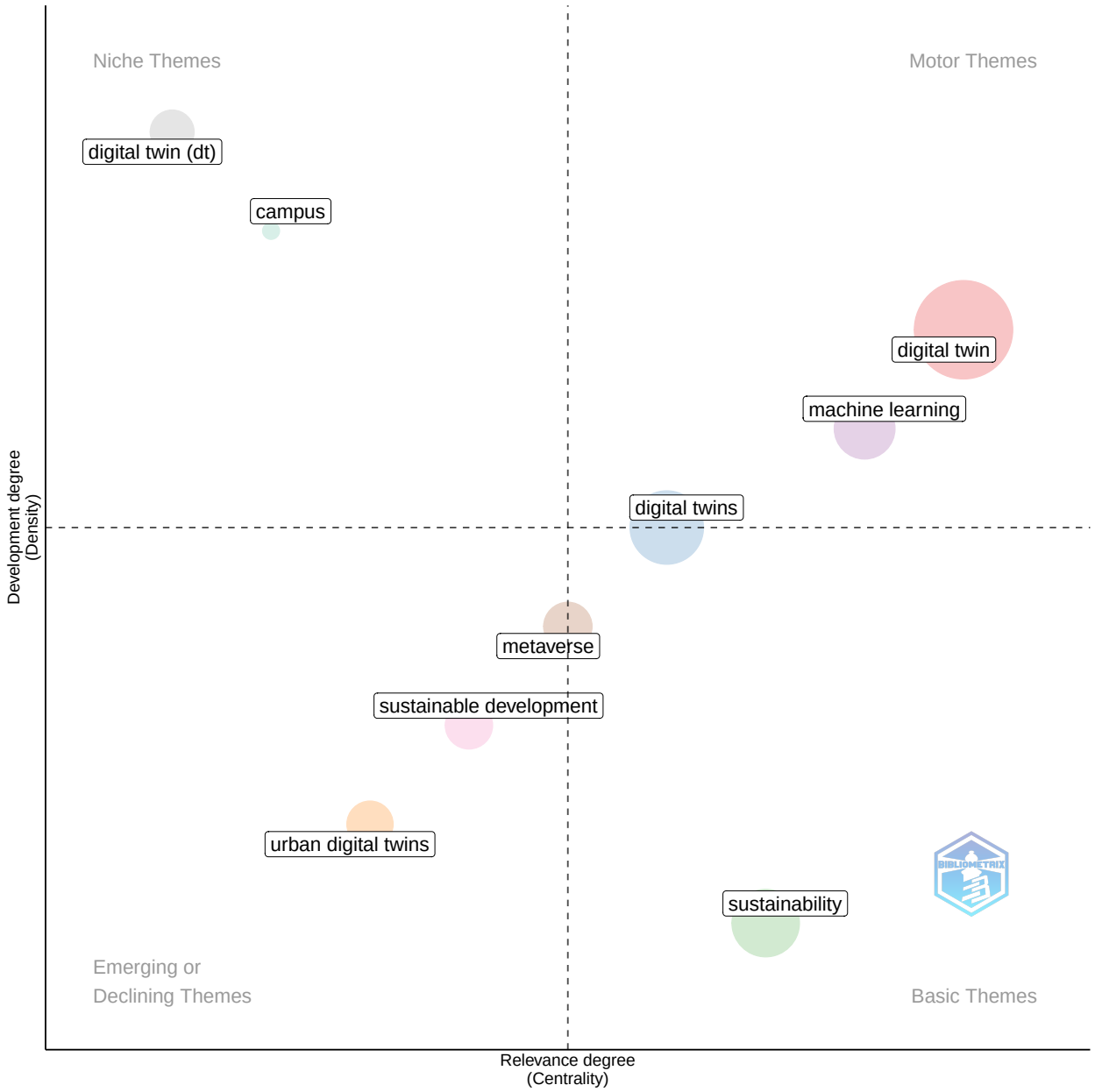

Figure 12: Thematic map — strategic positioning of research themes

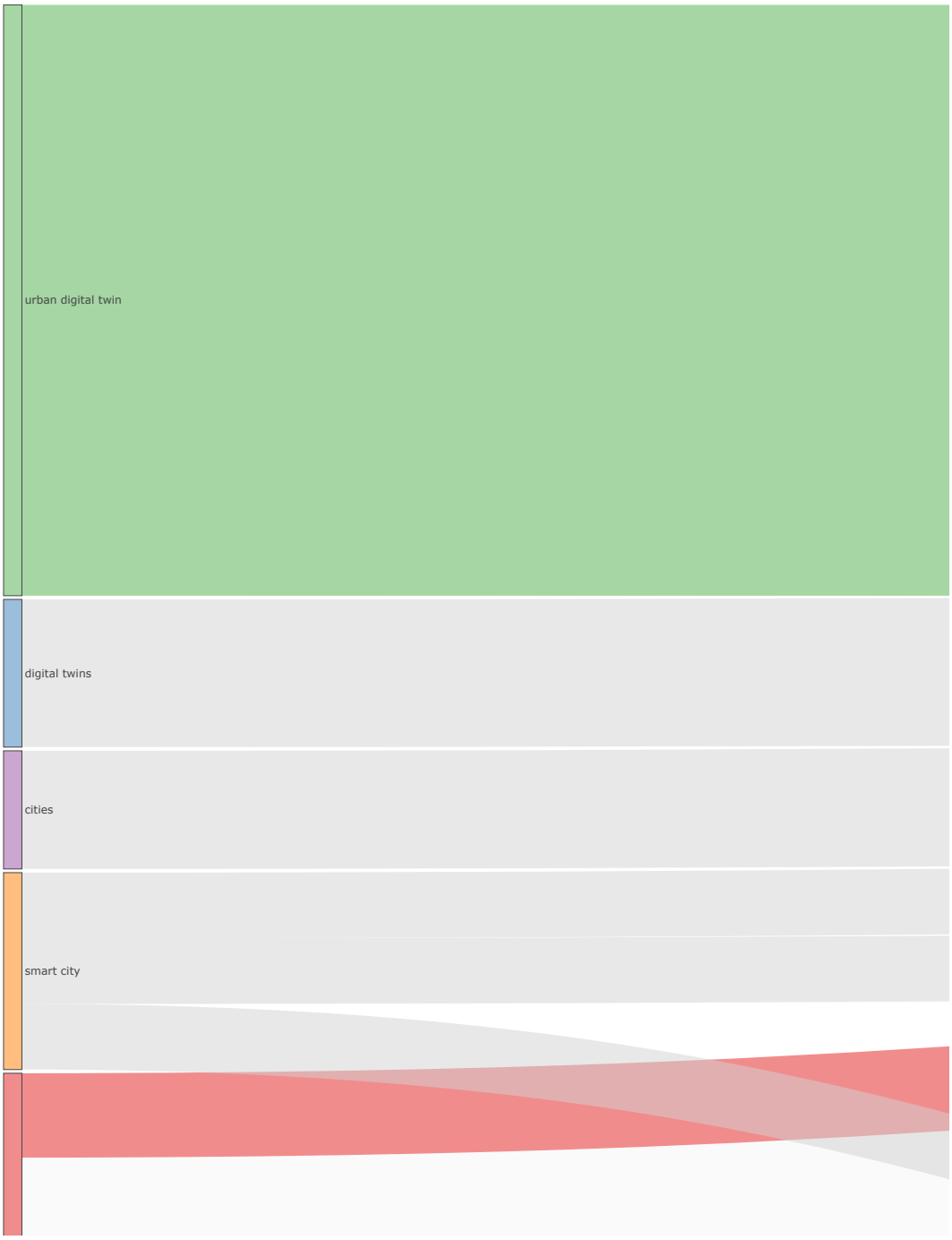

Figure 13: Thematic evolution over time

Table 10: Table 10 — Frequency of decision-support and public management terms in author keywords

	Term	Frequency
decision support	decision support	5
decision making	decision making	1
decision-making	decision-making	3
public management	public management	0
public administration	public administration	1
urban governance	urban governance	5
governance	governance	0
policy	policy	1
public policy	public policy	0
urban management	urban management	6
municipal	municipal	0
city management	city management	1

```
Clusters containing decision/management terms:

  'digital twin' → MOTOR (centrality: 9.000, density: 7.000)
  'machine learning' → MOTOR (centrality: 8.000, density: 6.000)
  'digital twin (dt)' → MOTOR (centrality: 1.000, density: 9.000)
```

18 Exportar Tabelas